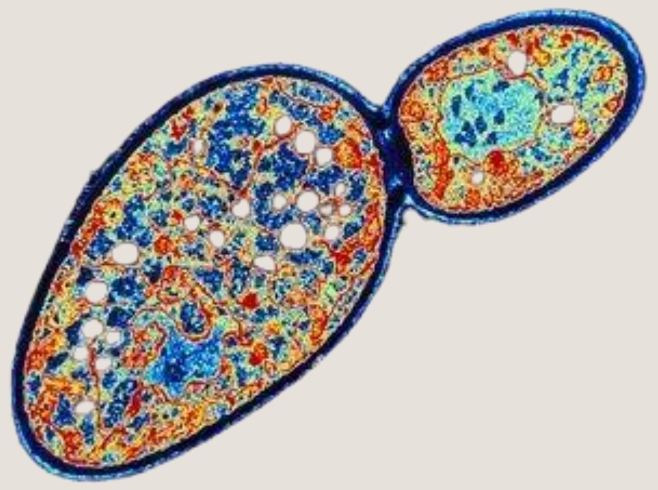
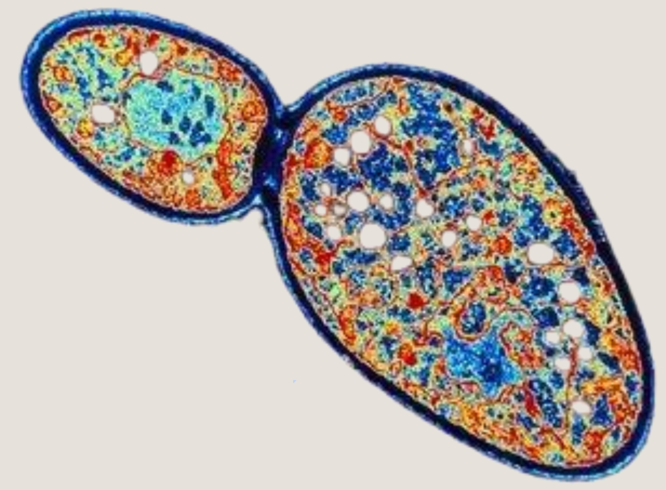


Crack, Purify, Detect : The Journey of Acid Phosphatase from *S.cerevisiae*



Introduction

Acid phosphatases are enzymes that catalyze the hydrolysis of phosphate esters in acidic conditions. They are present in many organisms and are important for phosphate metabolism. Previous extractions from wheat germ yielded poor results. This project aims to apply a multi-step biochemical protocol to extract, purify, and identify acid phosphatase (PAC) from *Saccharomyces cerevisiae*, using three main techniques: cell lysis and precipitation, ion-exchange chromatography, and SDS-PAGE electrophoresis.

1. Cell lysis and protein precipitation

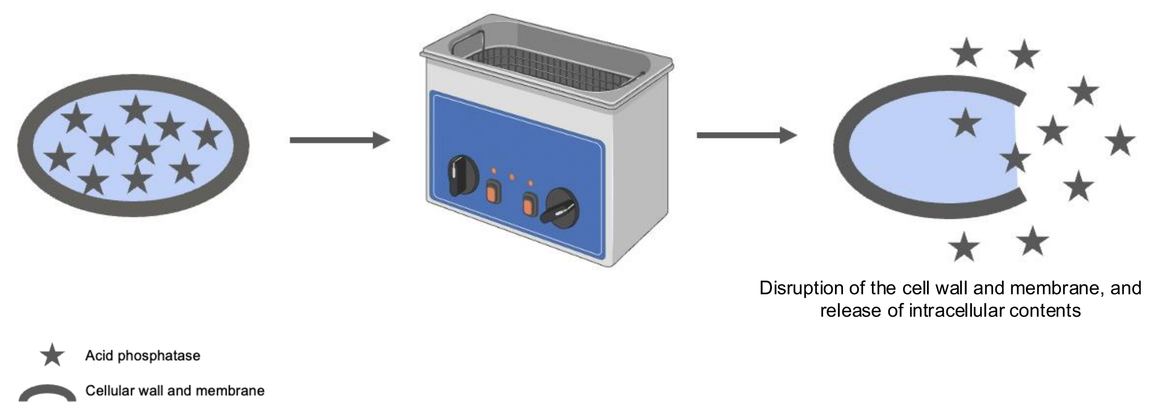
Role : This step partially purifies the sample by removing highly soluble or insoluble proteins, thereby enriching the target protein and making subsequent chromatography more effective.

Principle :

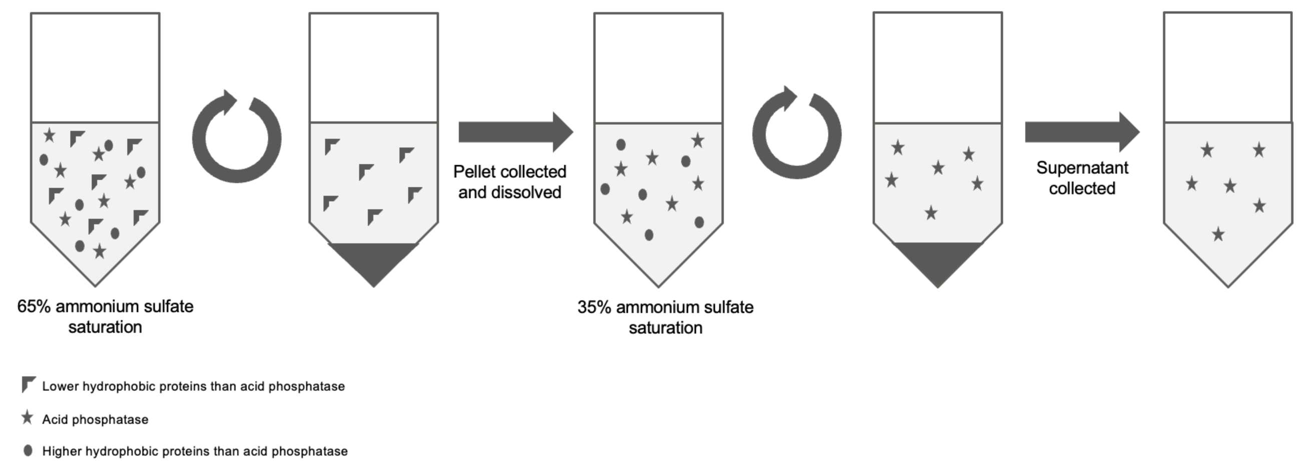
- Sonication breaks open yeast cells using high-frequency sound waves, releasing intracellular proteins such as PAC.
- Proteins are then separated from debris using centrifugation.
- Ammonium sulfate precipitation selectively precipitates proteins by reducing their solubility at specific saturation levels.

Technique Summary :

1. Yeast cells suspended in HEPES buffer are lysed by sonication (20 kHz, 3 min).



2. After centrifugation, the supernatant contains soluble proteins.
3. Proteins are precipitated at 65% then 35% ammonium sulfate saturation.



Zoom on the ammonium sulfate precipitations

Ammonium sulfate precipitation is a classical protein purification method based on "salting out". Proteins are less soluble at high salt concentrations because salt ions compete with protein molecules for water, leading to protein aggregation and precipitation. Different proteins precipitate at different saturation levels, depending on their surface charge and hydrophobicity.
After cell lysis and centrifugation, the supernatant containing soluble proteins underwent stepwise ammonium sulfate precipitation:
•At **65% saturation**, more hydrophobic or high molecular weight proteins are expected to precipitate. The pellet was collected and redissolved.
•A second precipitation at **35% saturation** was performed on the remaining supernatant, targeting proteins that remain more soluble at higher salt levels.
Each fraction may contain different subsets of proteins, including potentially the acid phosphatase, depending on its solubility properties. This method helps reduce sample complexity before chromatography.

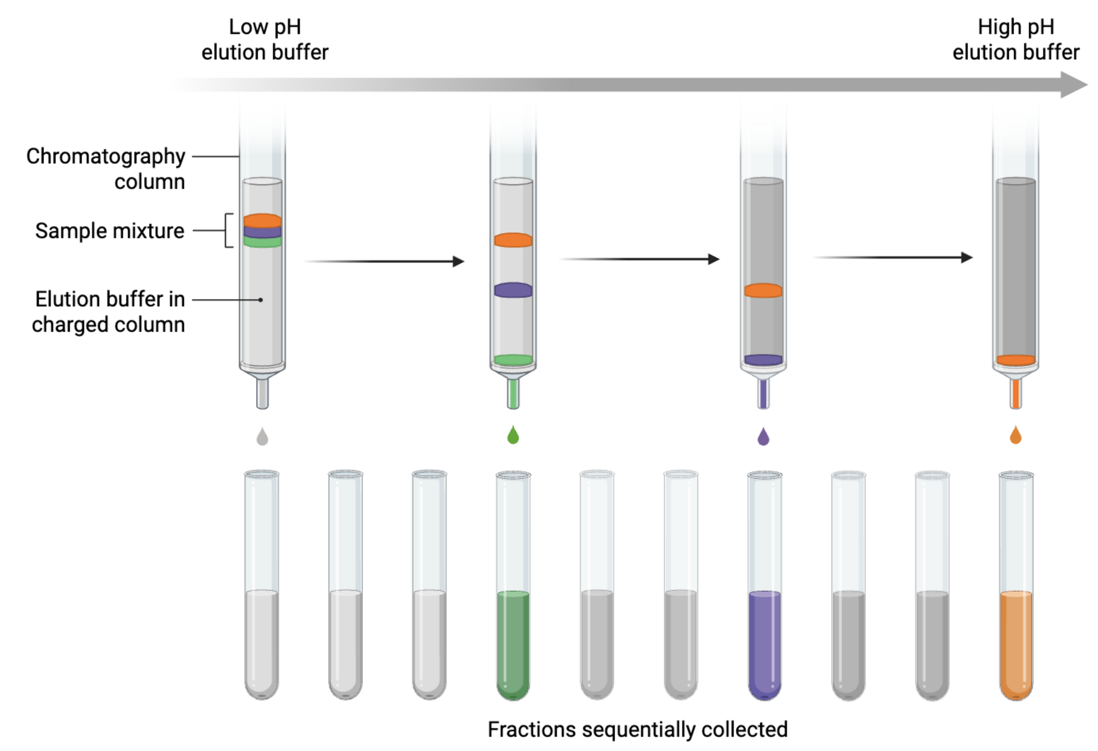
2. Ion-exchange Chromatography

Role : This simplified ion-exchange protocol was designed to elute groups of proteins according to their charge at different pH values. Acid phosphatase, which is expected to be less positively charged around pH 6–7, might elute at pH 6 or 13, depending on its isoelectric point.

Principle :
Ion-exchange chromatography separates proteins based on their net electric charge at different pH levels. Proteins interact with a charged resin in the column. Depending on the pH of the buffer, their charge changes, affecting how tightly they bind to the resin and when they elute.
In this experiment, a cation-exchange resin was used (negatively charged), which binds positively charged proteins. Elution was performed by successively increasing the pH.

Technique Summary :

1. A chromatography column was filled with cation-exchange resin pre-mixed with buffer.
2. The sample was added directly onto the column.
3. Three elution steps were performed:
    - Buffer at pH 4: most proteins are highly positively charged → bind tightly to the resin.
    - Buffer at pH 6: weakly bound proteins may start eluting.
    - Buffer at pH 13: proteins become negatively charged → elute completely from the resin.
4. The pH of each fraction was monitored using pH strips to identify which buffer was currently eluting.



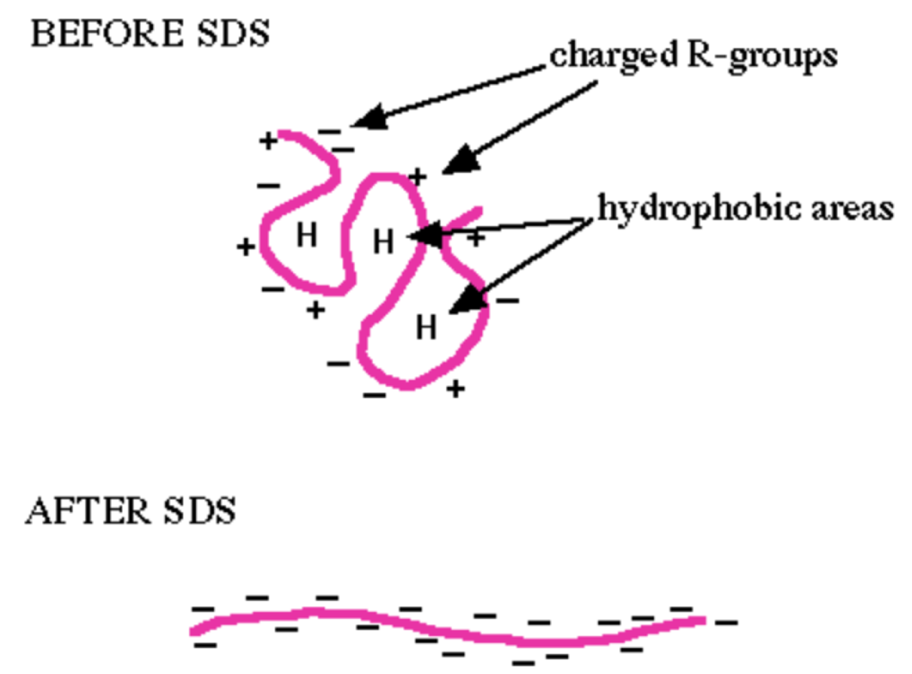
3. SDS - PAGE

Role : SDS-PAGE enables the qualitative assessment of protein purification by displaying band patterns. In this case, we expect to observe a distinct band near 60 kDa, corresponding to acid phosphatase, which should become more prominent and isolated in the later purification stages. The reduction in the number and intensity of non-specific bands across lanes would indicate increasing purity of the sample.

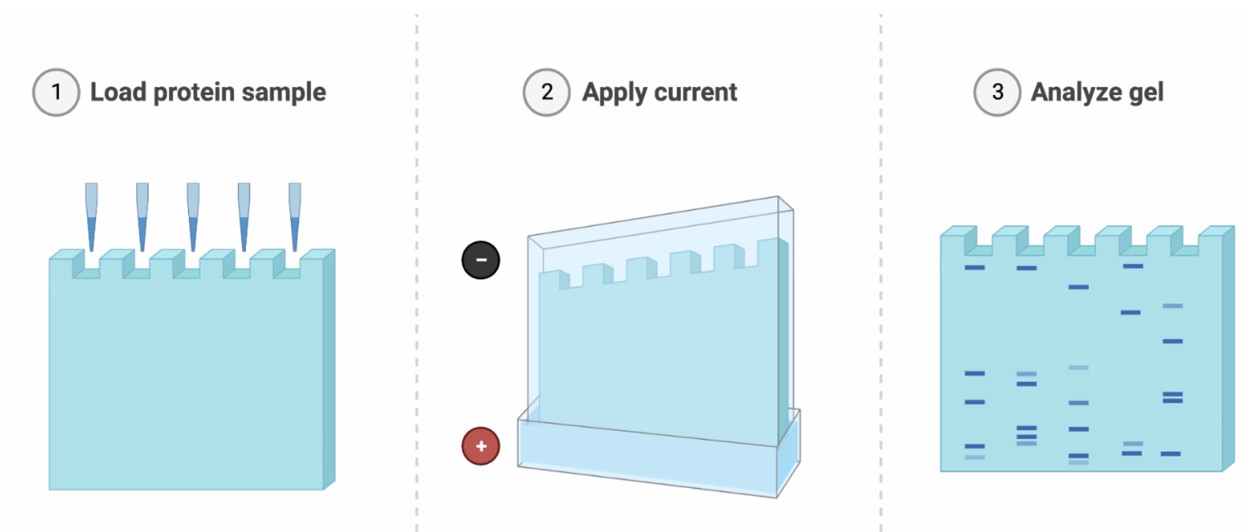
Principle :
SDS-PAGE (Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis) is a powerful technique used to separate proteins based on their molecular weight. In this method, proteins are first denatured by heat and reducing agents (such as β-mercaptoethanol), and coated with SDS, an anionic detergent that gives all proteins a uniform negative charge. As a result, protein migration through the polyacrylamide gel is influenced only by size, not by charge or shape.

Technique Summary :

1. Protein samples from different purification steps were mixed with Laemmli buffer, containing SDS and β-mercaptoethanol, and heated at 97°C to denature them completely.



2. Samples were then loaded into wells of a 10% polyacrylamide gel, along with a molecular weight marker.
3. Under an electric field (140 V), proteins migrated toward the anode (+), with smaller proteins moving faster through the gel matrix.
4. After migration, the gel was stained using Coomassie Brilliant Blue, which binds to proteins and reveals distinct bands.
5. Each lane corresponds to a sample taken at a different step of the protocol (crude extract, post-sonication, after centrifugation, after precipitation, after chromatography), allowing a comparative visualization of protein composition and purification progress.



Conclusion

This experimental approach aimed to isolate and identify acid phosphatase from *Saccharomyces cerevisiae* using sequential biochemical techniques.
Although no direct enzymatic results were obtained, the protocol should allow progressive enrichment of the target protein. The modified ion-exchange chromatography, based on stepwise elution at pH 4, 6 and 13, was designed to separate proteins by their changing charge. Acid phosphatase, which is likely less positively charged at neutral to basic pH, may have eluted in the pH 6 or pH 13 fractions. The final SDS-PAGE should help visualize a specific protein band around 60 kDa, supporting the hypothesis of successful purification.
However, the method also presents limitations: the stepwise elution may lack precision, and protein loss during precipitation and handling is likely. To confirm identity and enzymatic activity, further analysis such as activity assays or immunodetection would be necessary in future work.

Contact

Amandine Arliaud
amandinearliaud@gmail.com
0619822878
GBIO
IUT DE TOULON